CASE STUDY

Police Station • Salt, Spain

POLICE STATION

Spatial Intelligence

PHASE 02 · GROUND FLOOR ANALYSIS

This study applies graph-based spatial analysis to the ground floor of a police station, treating architectural space as a network of nodes and adjacencies. Using topologicPy and graph theory, Phase 02 examines centrality, community structure, and isovist visibility — revealing the spatial intelligence embedded in the organisation of the building.

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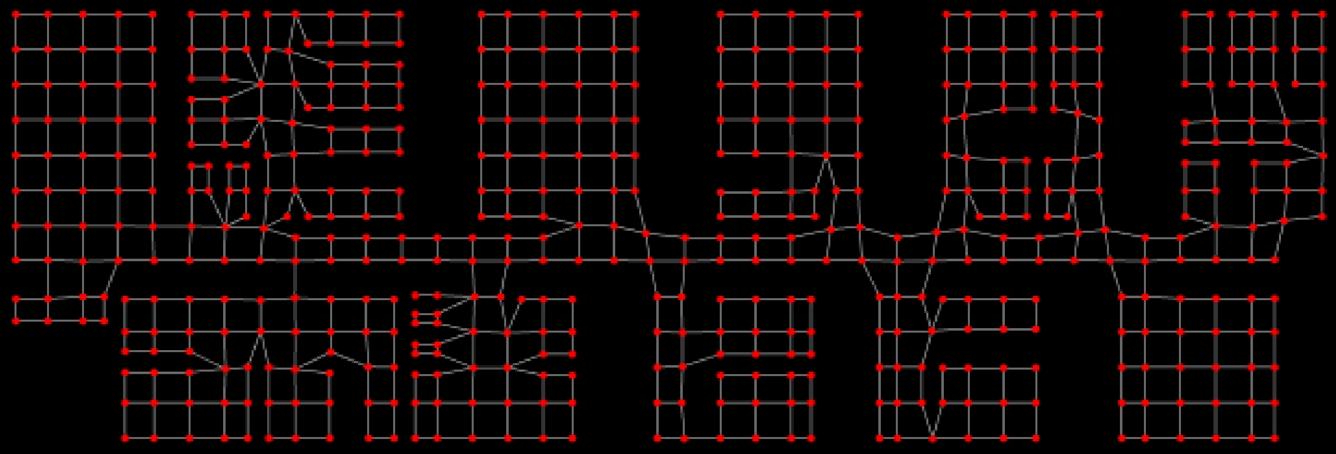
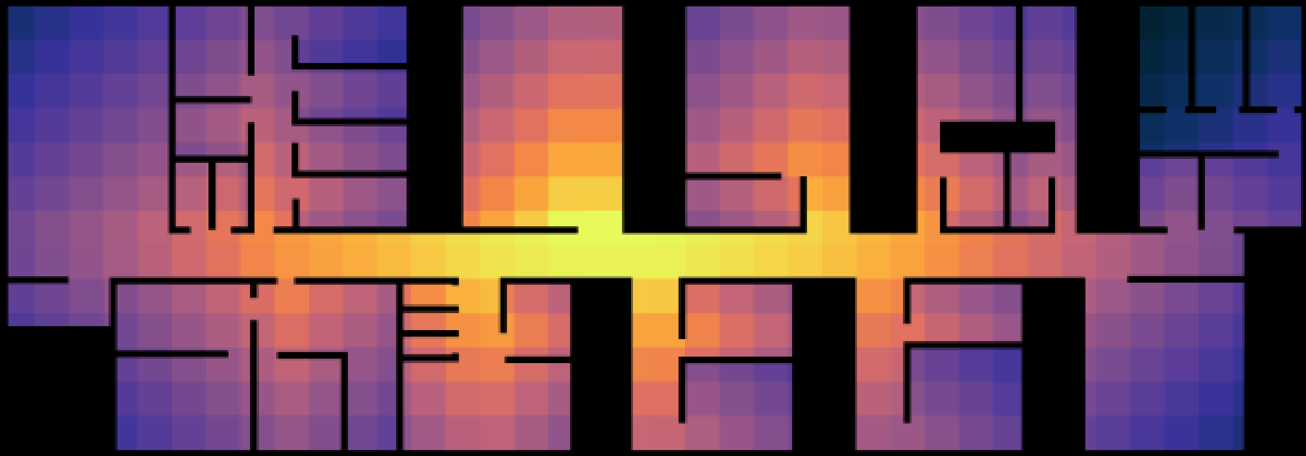
grapML · Wassim Jabi



01

ANALYSIS GRAPH

The ground floor plan is discretized into a 2×2m grid, each cell becoming a node and shared edges becoming connections. This graph encodes spatial proximity as

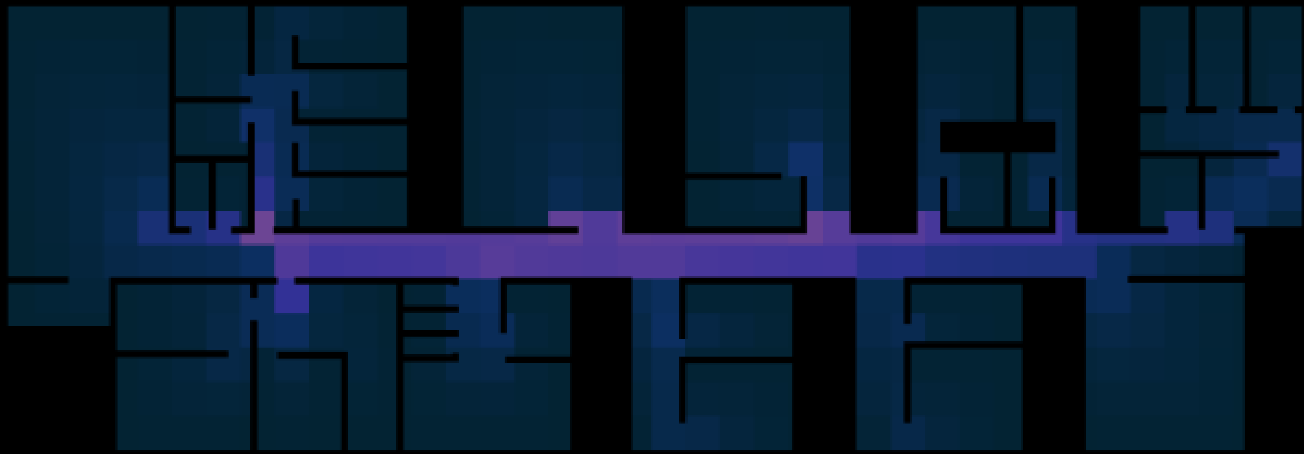


02

CLOSENESS CENTRALITY

· INTEGRATION

The main corridor is the shallowest space in the building — reachable from any point in the fewest steps. Peripheral wings read as topologically isolated. Warm



03

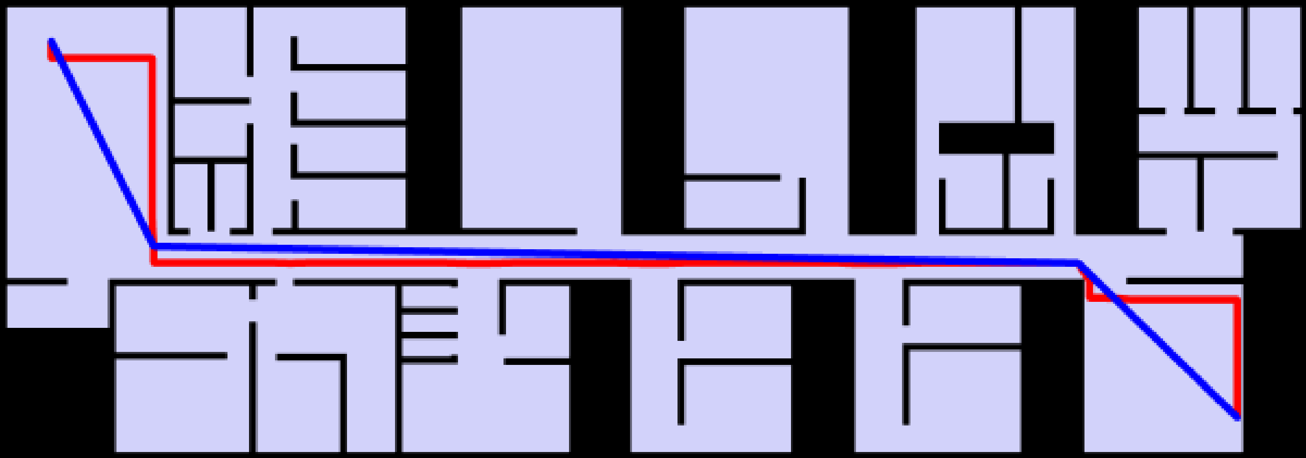
BETWEENNESS CENTRALITY

· CHOICE

A single east–west spine concentrates almost all routing choice. Nearly every shortest path between any two spaces passes through the corridor — one structural bottleneck

Betweenness Centrality

LOW CHOICE HIGH CHOICE



SHORTEST PATH

Cross-building traversal, NW to SE corner. The graph path follows cell boundaries; geometric straightening cuts across open space, reducing travel distance by 11.3%.

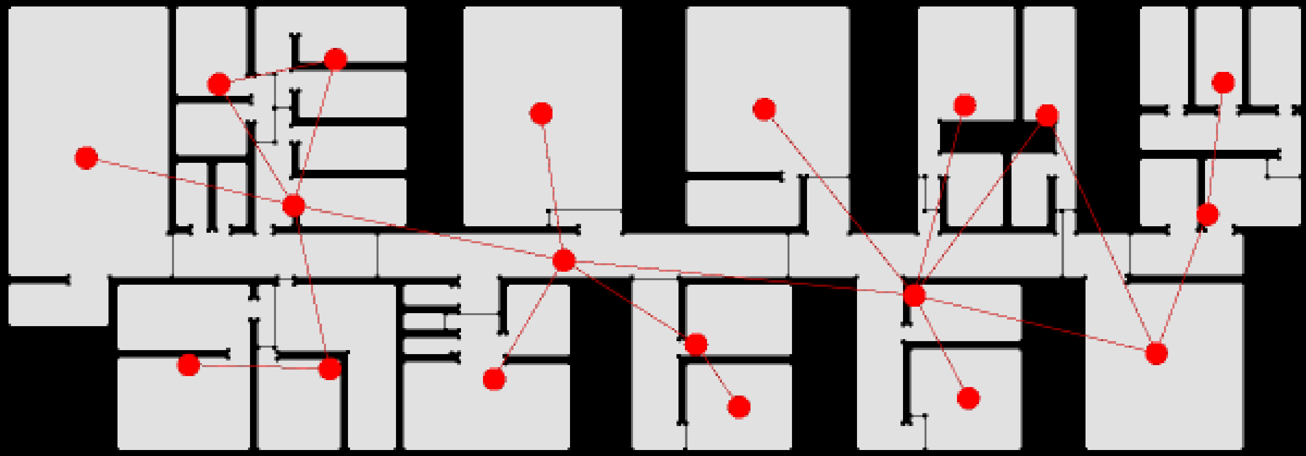
Graph path — 90.65 units

Straightened — 80.41 units (-11.3%)

COMMUNITY PARTITION

6 spatially coherent zones emerge — clusters of cells more strongly connected internally than to their neighbours. These map onto the functional wings of the

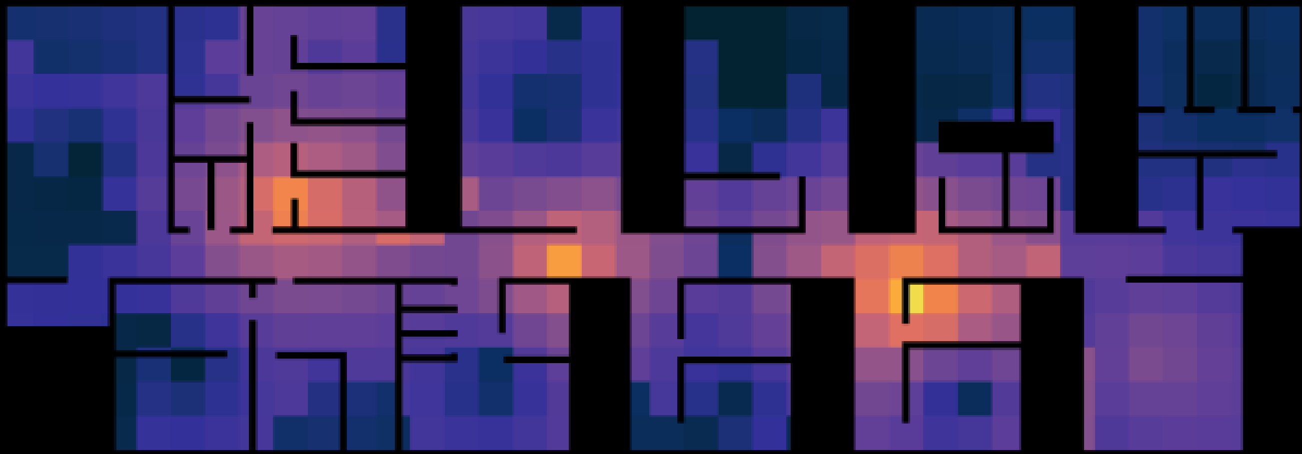
Zone 1 Zone 2 Zone 3
Zone 4 Zone 5 Zone 6



06

COMMUNITY GRAPH

Communities abstracted to single nodes reveal the inter-zone topology. The corridor community acts as the hub with the highest degree. Its removal would fragment



07

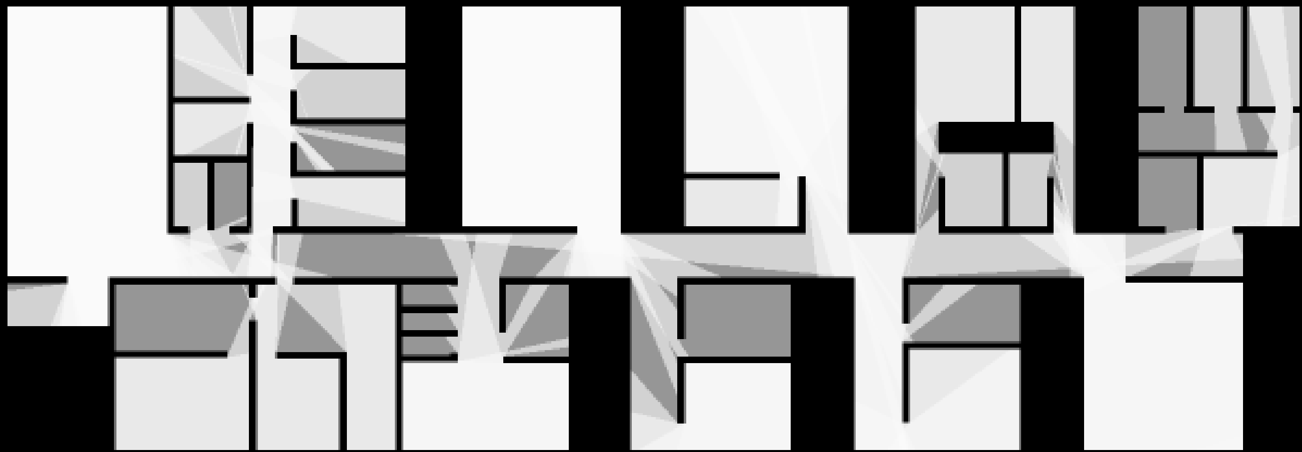
DEGREE CENTRALITY

· COMMUNITY SCALE

Degree centrality interpolated from community nodes back to the grid. Junction zones at the boundary between corridor and wings emerge as the most topologically

Degree Centrality

LOW HIGH



08

ISOVIST VIEWPOINTS

61 viewpoints sampled at 5m intervals across the floor plan. Each isovist polygon captures the total area visible from that position. Visibility ranges from 6 to

Viewpoint (61 total)

Isovist (visibility polygon)

09

ISOVIST VISIBILITY MAP

Visibility scores interpolated across the full floor plan. Corridor junctions and open spaces are the most exposed. Subdivided rooms and dead-end wings register the

Visibility Score

HIDDEN EXPOSED